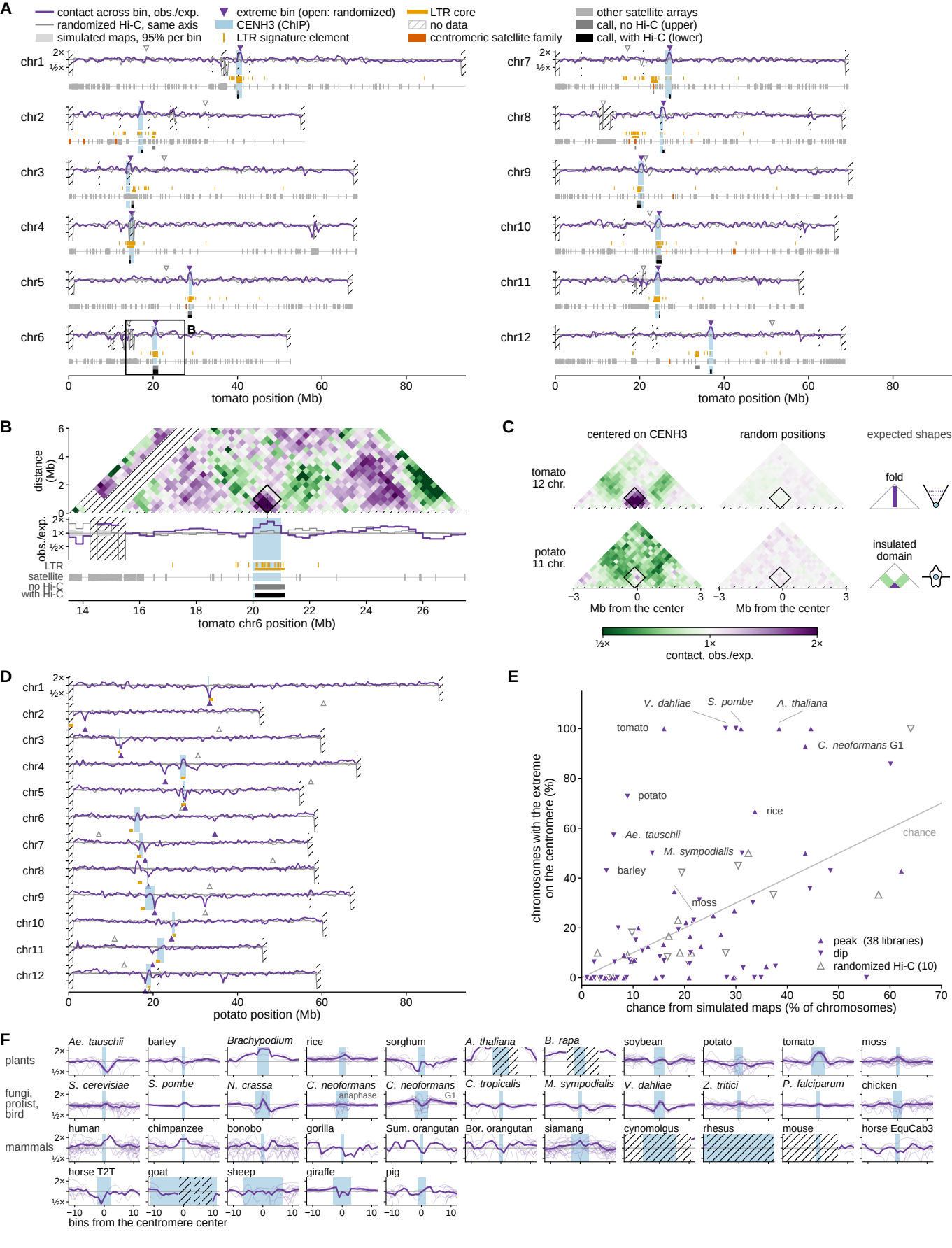

A
contact across bin, obs./exp.
randomized Hi-C, same axis
simulated maps, 95% per bin
extreme bin (open: randomized)
CENH3 (ChIP)
LTR signature element
LTR core
no data
centromeric satellite family
other satellite arrays
call, no Hi-C (upper)
call, with Hi-C (lower)
chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
tomato position (Mb)
B
distance (Mb)
obs./exp.
LTR
satellite
no Hi-C
with Hi-C
tomato chr6 position (Mb)
C
centered on CENH3
random positions
expected shapes
tomato 12 chr.
potato 11 chr.
fold
insulated domain
Mb from the center
contact, obs./exp.
D
potato position (Mb)
E
chromosomes with the extreme on the centromere (%)
chance from simulated maps (% of chromosomes)
tomato
potato
rice
barley
moss
V. dahliae
S. pombe
A. thaliana
C. neoformans G1
Ae. tauschii
M. sympodialis
chance
peak (38 libraries)
dip
randomized Hi-C (10)
F
plants
fungi, protist, bird
mammals
Ae. tauschii
barley
Brachypodium
rice
sorghum
A. thaliana
B. rapa
soybean
potato
tomato
moss
S. cerevisiae
S. pombe
N. crassa
C. neoformans anaphase
C. neoformans G1
C. tropicalis
M. sympodialis
V. dahliae
Z. tritici
P. falciparum
chicken
human
chimpanzee
bonobo
gorilla
Sum. orangutan
Bor. orangutan
siamang
cynomolgus
rhesus
mouse
horse EquCab3
horse T2T
goat
sheep
giraffe
pig
bins from the centromere center

The two sides of every tomato centromere contact each other about twice as often as equally distant DNA elsewhere, so a contact map alone marks centromeres that no repeat marks. Potato shows a deficit instead, and the signal recurs in some plants and fungi but in no mammal.

A On all 12 tomato chromosomes, contact between the two sides of each point peaks at the CENH3 domain; randomized Hi-C and simulated maps show no such peak. The centromeric satellite and the retrotransposons that mark centromeres often lie elsewhere, so calls from the genome and its transposons alone miss four centromeres, two of them confidently (upper row); with Hi-C, all 12 are placed (lower row).

B Tomato chr6 (box in A) as a contact map: the excess is a compact patch of contact between the two sides of the centromere, directly above it. The diamond holds the contacts that make the profile value beneath.

C Averaged over chromosomes, tomato centromeres gain contact across the centromere and lose contact with the rest of the chromosome, more like a small insulated domain than arms folded together; potato centromeres lose contact instead, and random positions stay flat. The cartoons are expectations, and the mechanism is a hypothesis.

D In potato the same measure dips at most centromeres instead of peaking (triangles: lowest bin; orange bar: the retrotransposon core). Here Hi-C adds no centromere: calls place 9 of 11 with it and 10 without.

E In 11 of 37 genomes with Hi-C and a centromere map, the peak or the dip falls on the centromere more often than chance (line): 7 of 11 plants, 4 of 8 fungi, and none of the 16 mammals, *Plasmodium* or chicken. Each point is one Hi-C library; *C. neoformans* has two and counts once, as its G1 library beats chance and its anaphase library does not. Open marks: randomized Hi-C of 10 of these genomes stays near chance; its one outlier, a dip in moss, is what chance alone yields among 20 tries about one time in ten.

F The average profile around the centromeres of each genome (faint lines: single chromosomes; blue: centromere width; hatched: no data) shows which genomes peak, which dip and which stay flat. Hi-C bins: 250 kb; 2 Mb in *Ae. tauschii* and barley; 500 kb in sorghum, human and mouse; 50 kb in *S. pombe*, *N. crassa*, *C. tropicalis* and *V. dahliae*; 10 kb in *S. cerevisiae*, *C. neoformans*, *M. sympodialis*, *Z. tritici* and *P. falciparum*.

Script: docs/img/src/benchmark_cis.py · data: the cenfuse 0.16.0 benchmark runs (not in the repository) · 2026-11-14